from A to B contribute: it is as if the electron, in order to go from A to B, passed ‘through all possible trajectories’, or, in other words, unfurled into a cloud in order then to converge mysteriously on point B, where it collides again with something else (figure 4.7).

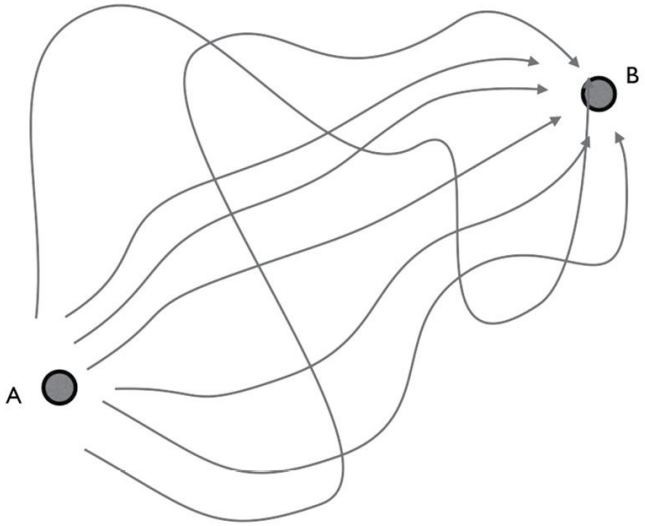


Figure 4.7 In order to move from A to B an electron behaves as if passing through *all* possible trajectories.

This technique for computing the probability of a quantum event is called Feynman’s sum over paths,[fn27] and we shall see that it plays a role in quantum gravity.

Quanta 3: Reality is relational

The third discovery about the world articulated by quantum mechanics is the most profound and difficult – and one which was not anticipated by the atomism of antiquity.

The theory does not describe things as they *are*: it describes how things *occur* and how they *interact with each other*. It doesn’t describe where there is a particle but how the particle *shows itself to others*. The world of existent things is reduced to a realm of possible interactions. Reality is reduced to interaction. Reality is reduced to relation.[4]

In a certain sense, this is just an extension of relativity, albeit a radical one. Aristotle was first to emphasize that we only perceive *relative* speed. On a ship, for example, we talk of our speed relative to the ship; on land, relative to the Earth. Galileo understood that this is the reason why the Earth can move with respect to the Sun without us feeling the movement. Speed is not a property of an object on its own: it is the property of the motion of an object *with respect to another object*. Einstein extended the notion of relativity to